
DIGITAL DISCOM & DIGITIZED CONSUMER

India Smart Utility Week 2026



POWER FOUNDATION OF INDIA

March, 2026

BACKGROUND

Power Foundation of India is a registered Society under the aegis of MoP

To support the Ministry through regulatory and research support, advocacy and outreach programs

Evidence-based policy research , Regulatory and analytical support, Capacity building, awareness campaigns, and institutional strengthening, Supporting India's Energy Transition, Vision Viksit Bharat @2047 and Net-Zero 2070 goals

The Society is funded through annual membership fee from leading power sector CPSEs

GOVERNANCE STRUCTURE OF PFI

Steering Council

To guide the Foundation in carrying out its activities, setting of agendas and to give suggestions on behalf of Government

Chairman - Hon'ble Minister of Power
Members – Secretaries of Various Ministries
CEO – Niti Aayog
Chairperson – CEA
CMDs of Power CPSEs
DG- ISA and PFI

Executive Committee

Chairman - Director General, PFI
Members – Addl. Secretary, M/o Power
Chairperson – CEA/ Nominee Representative
CMDs / Nominee Representative of Power CPSEs
DG / Nominee Representative of CPRI, NPTI and CSIR
Convenor- Executive Director, PFI

Governing Body

To manage and administer all affairs of the society

Representatives of Power sector CPSEs

MoUs Signed



Key Activities

1. Studies

In-house
Institutes

2. Party before SERCs

DISCOMs Tariff Petitions
Draft Regulations

3. Chair Professorship

IISC Bangalore
IIT Kanpur
IIM Raipur (under progress)
(Centre of Energy Regulations)

4. Knowledge Partner

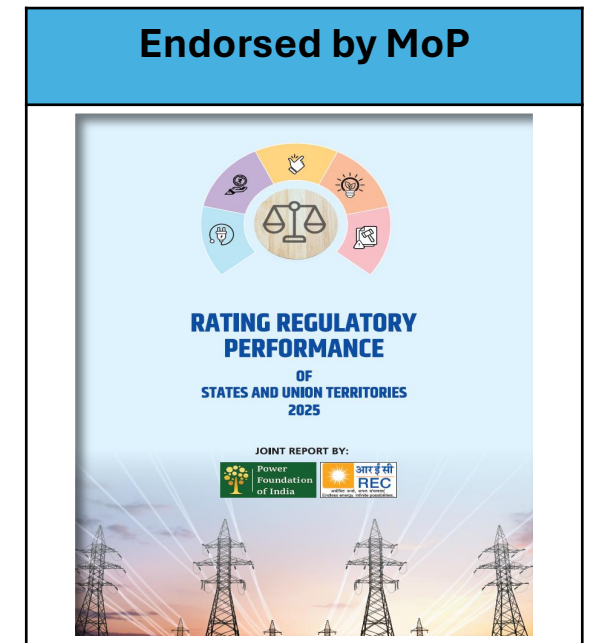
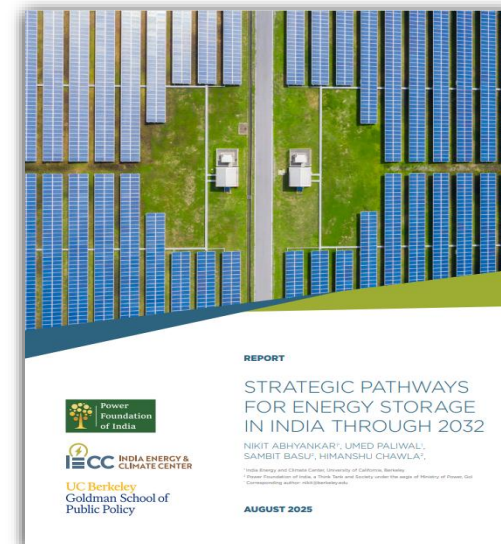
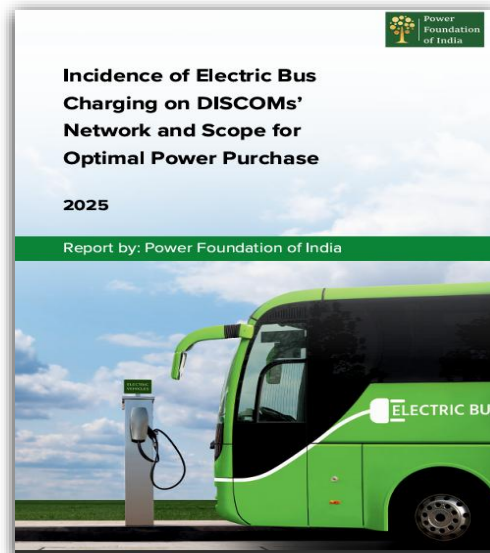
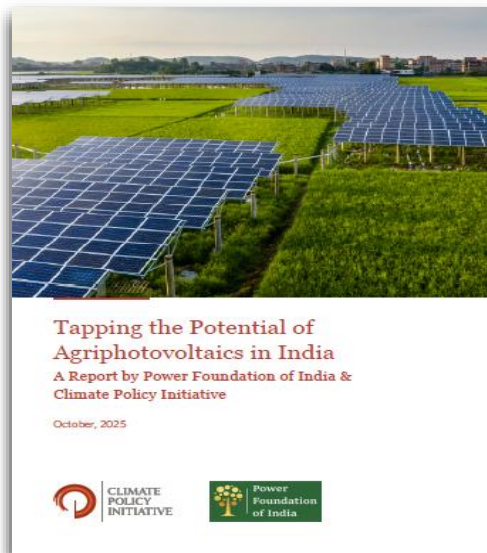
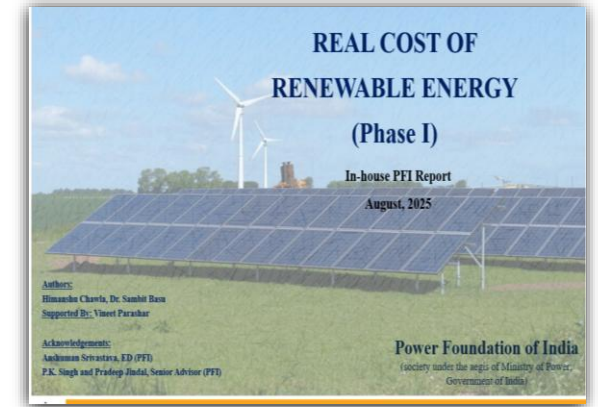
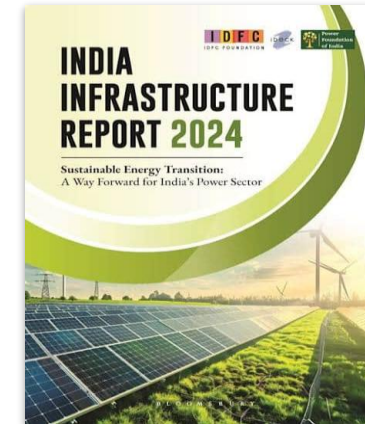
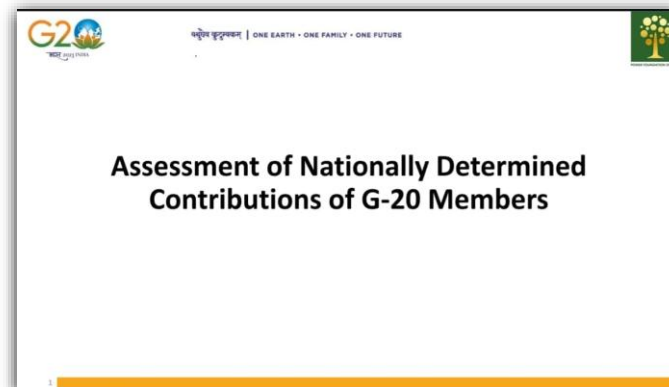
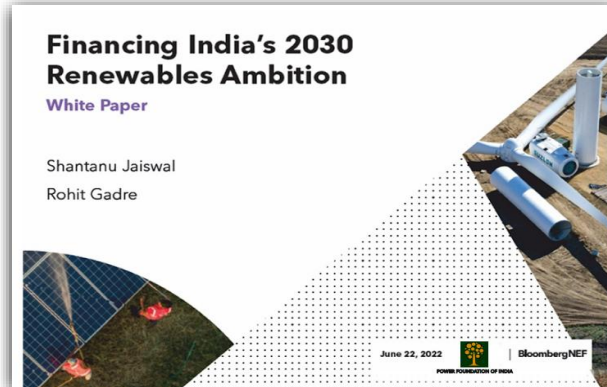
DUM 2025
ISUW 2026

5. Regional Regulatory Workshops

6. Concept Notes for MoP, Snapshots, Articles

PFI: Studies Completed

Evidence-based Research

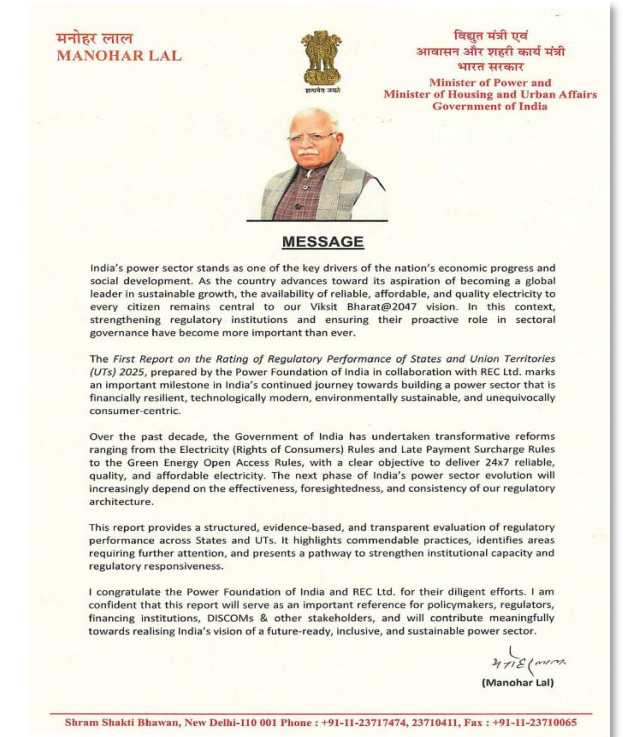
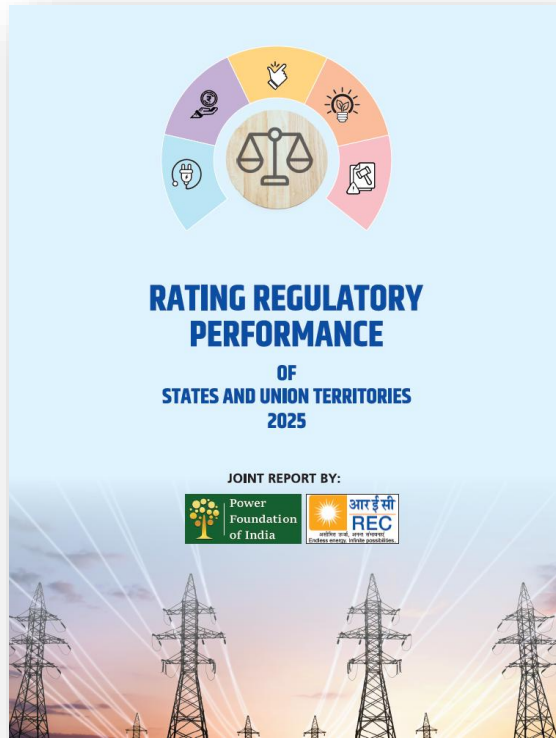


PFI: Studies Completed

Endorsed by MoP

RATING REGULATORY PERFORMANCE OF STATES AND UNION TERRITORIES

- First of its kind Report in Power Sector - Rating all 36 States & UTs
- Five Key Parameters – Resource Adequacy, Financial Viability of DISCOMs, Ease of Living / Doing Business, Energy Transition & Regulatory Governance
- **Consultative Process Followed**
- **Hon'ble Union Minister (Power and Housing & Urban Affairs) has granted approval, Report published**



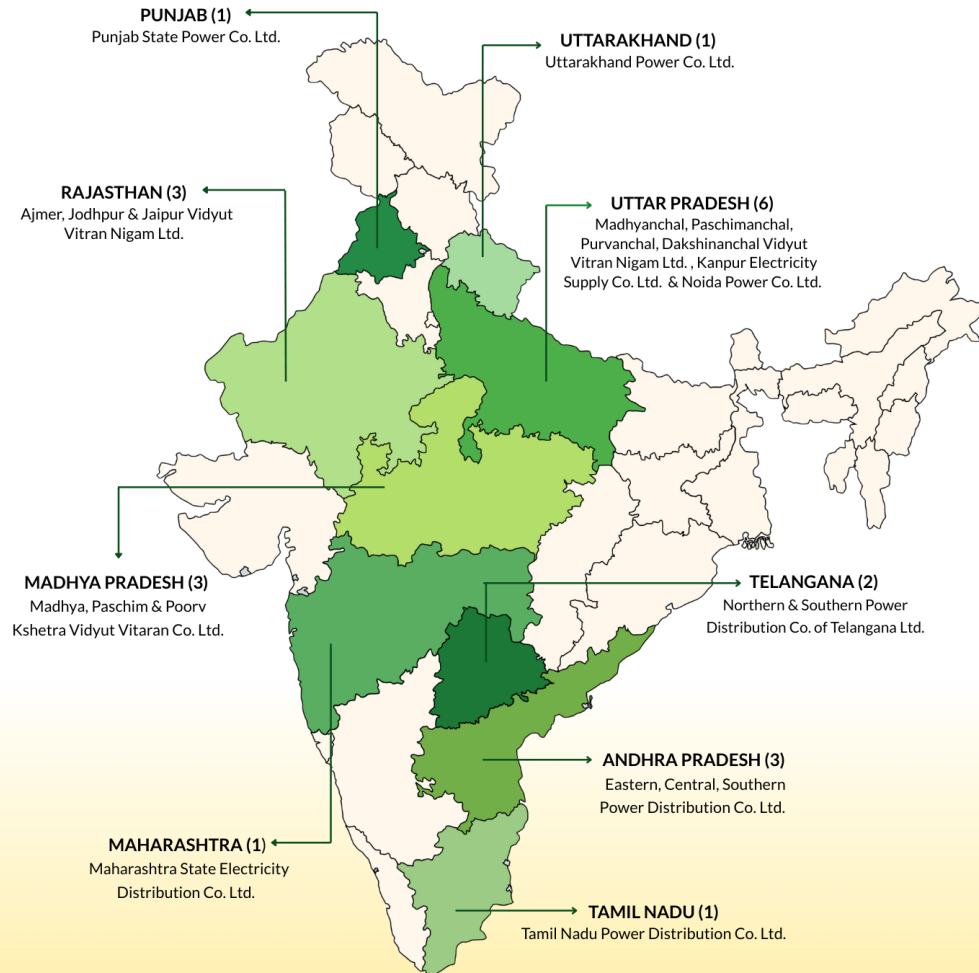
PFI: Ongoing Research Studies

Ongoing Studies		Institution
1	Accelerating adoption of Demand Side Management	Prayas
2	Cost of Variable Renewable Energy Integration (Phase-II)	IIT Kanpur
3	Policy Interventions for Net Zero – Lessons from Global Experiences	IIT Kanpur
4	Root Cause of increasing Case Load petitioned with APTEL	IIM Ahmedabad
5	Challenges faced by DISCOMs in meeting RCO from contract with REIA	IIM Ahmedabad
6	Impact of Electricity Subsidies on the Indian Economy	IIM Raipur

PFI PARTY BEFORE SERCs - DISCOMs' Tariff Petitions

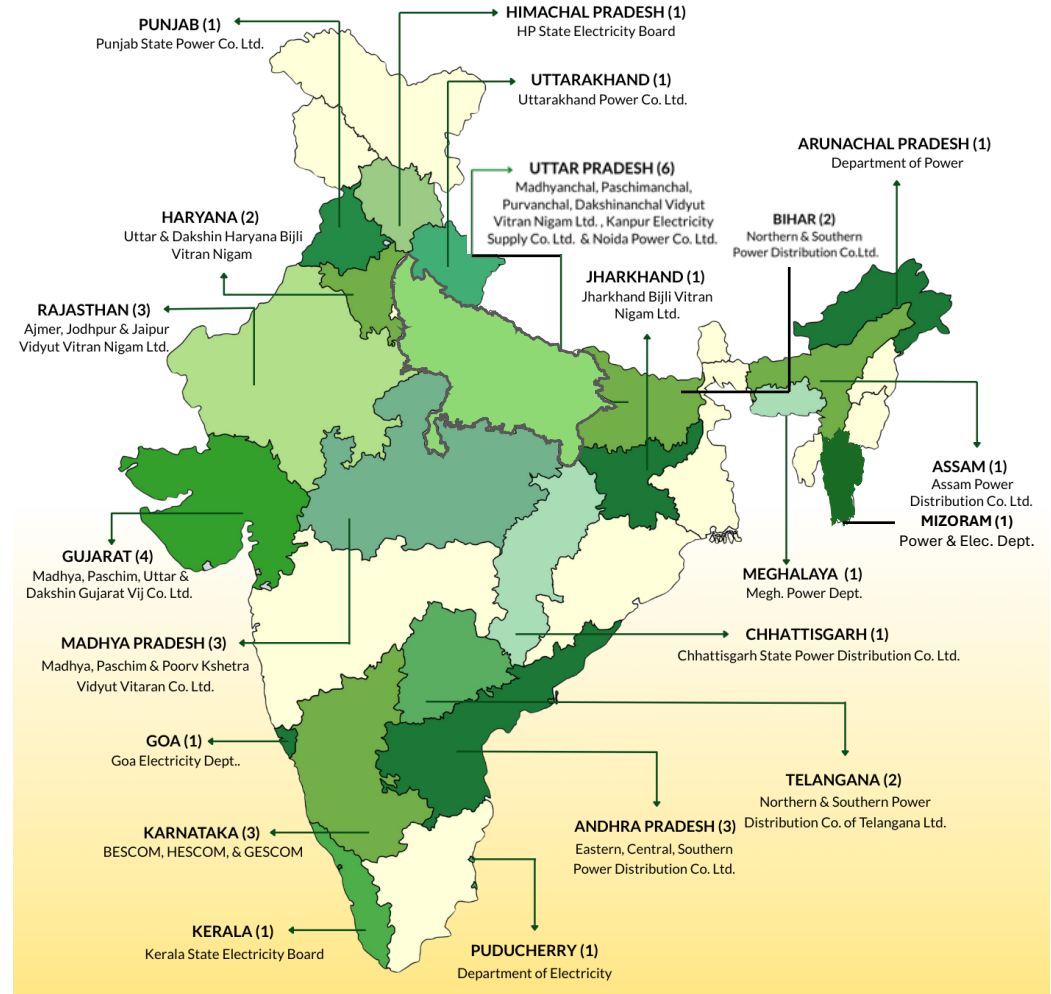
21 critical DISCOMs across 9 States, 18 True-Up Petitions for FY 2023-24, 22 ARR & Tariff Petitions for FY 2025-26

Comments Submitted in 2024-25



40 critical DISCOMs across 21 States, 39 True-Up Petitions for FY 2024-25, 34 ARR & Tariff Petitions for FY 2026-27

Comments Submitted in 2025-26



Data Analytics

PFI Energy Dashboard



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Demand

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Generation

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Transmission

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Distribution

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Supply

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Environment

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Economy & Demography

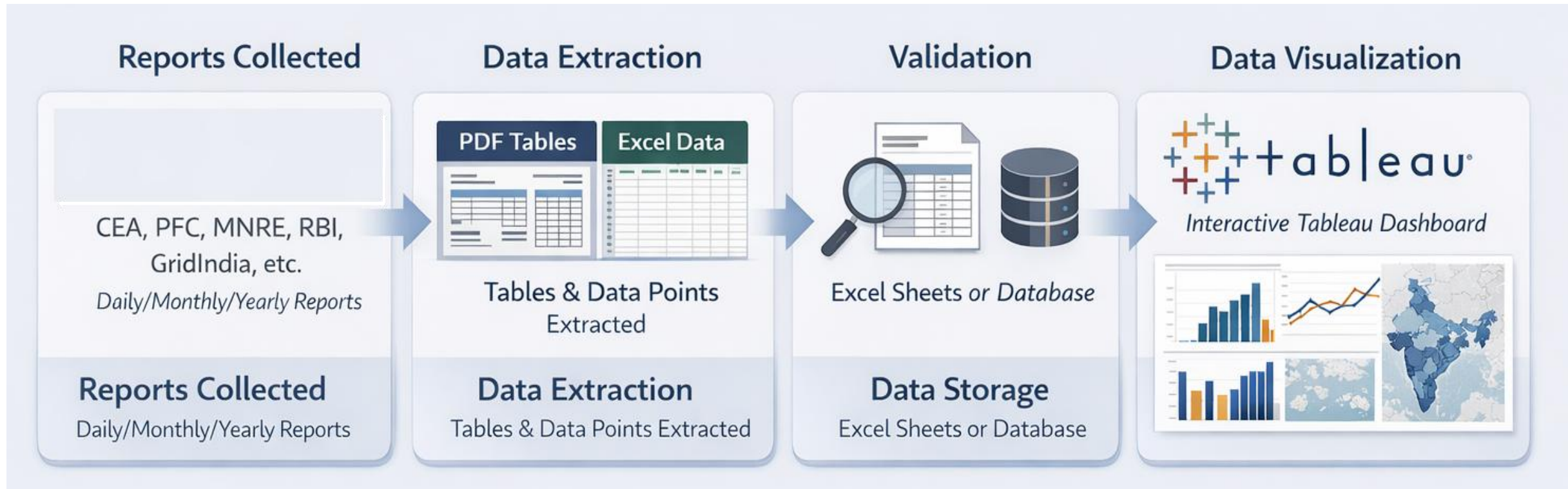
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Analytics

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Data Analytics

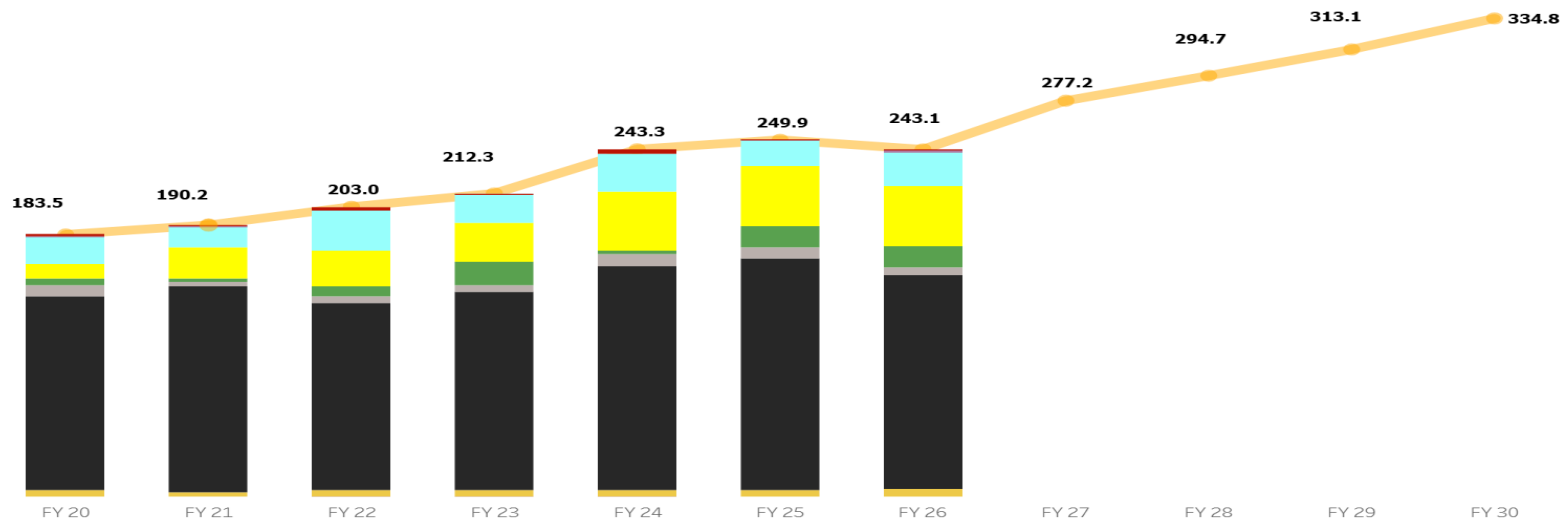
PFI Energy Dashboard



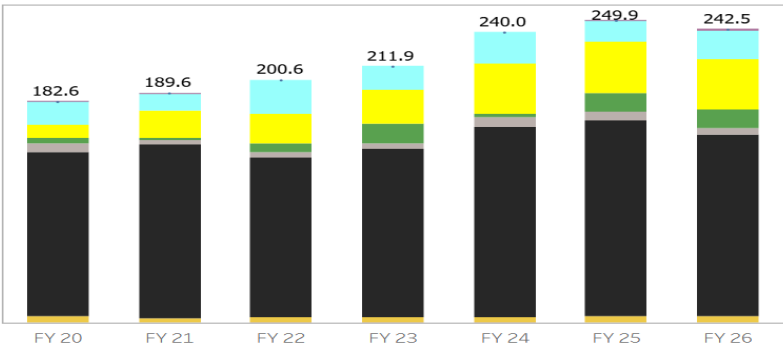
Data Analytics

Peak Demand and Source-Wise Contribution

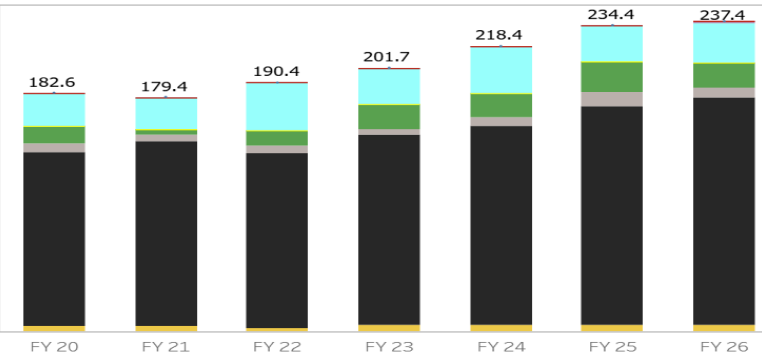
All Figures in GW



Contribution in Solar Hours



Contribution in Non-Solar Hours



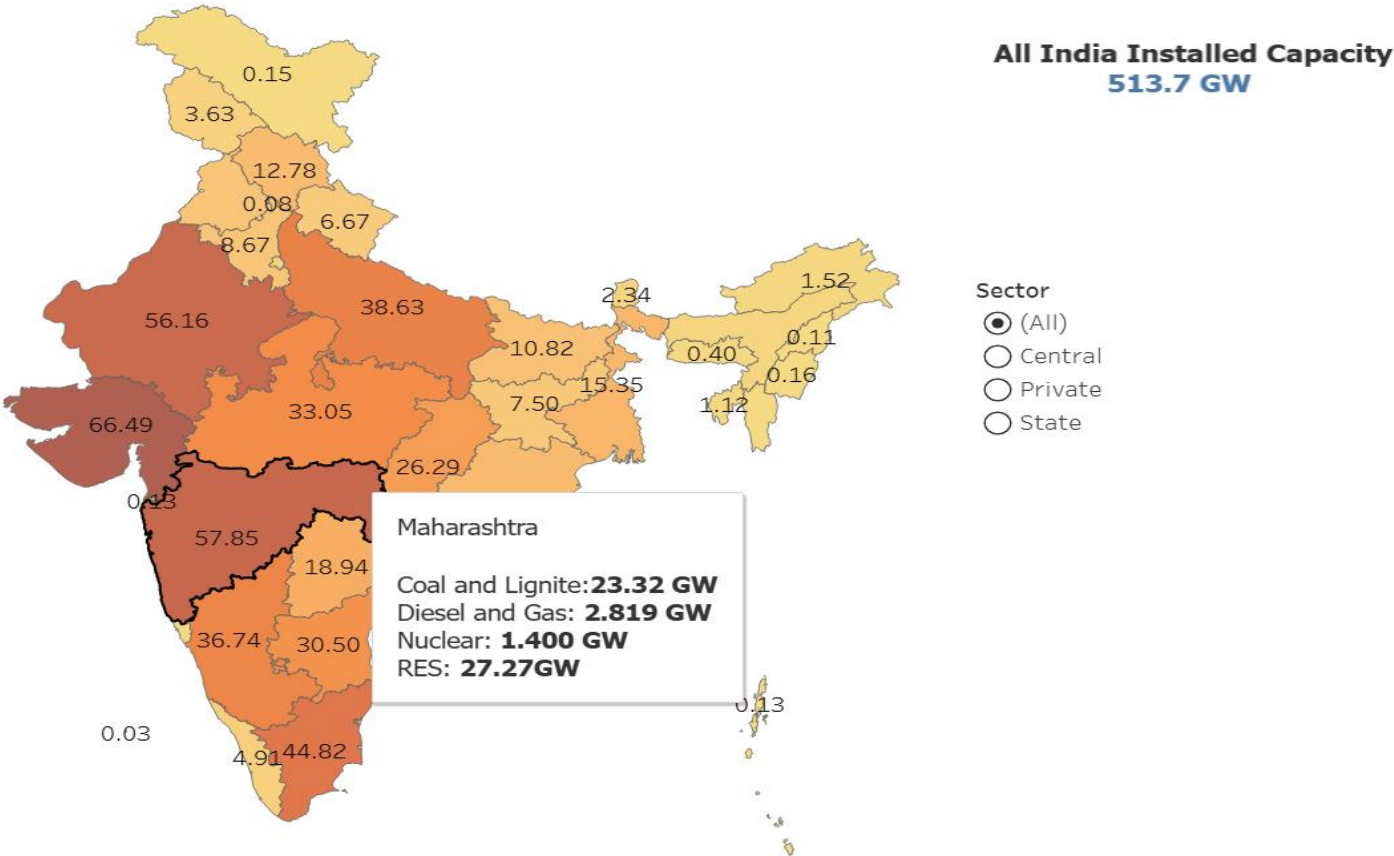
Note: FY 2027 to FY 2032 Data taken from CEA EPS Report
Source: NLDC

State-wise Demand

Data Analytics

State-wise Installed Capacity

(All figures in GW) As on 30 Dec 2025



RES includes Small Hydro, Solar, Wind Power and Bio-Power (does not include Large Hydro)
 Source: NPP. (Nov 2025). Installed Capacity- Monthly, accessed 09 Feb 2026

Communication and Outreach

Strengthens the Foundation's objectives through **awareness campaigns, conference and workshops**, and high-quality **multimedia content** for effective knowledge dissemination.

Awareness Campaigns

Rights of (Electricity) Consumers

- Awareness on Electricity Rights of Consumers Rules, 2020
- Pan-India digital campaign + outdoor outreach
- Multilingual

Mission LiFE

- Promotion of sustainable lifestyles
- 7 conferences + 1 national exhibition

Power Sector Achievements

Outdoor & selfie booths across Delhi NCR

Conferences and Workshops

Dissemination Workshop: *Electric Bus Charging on DISCOMs' Network* report (Jan 2026)

India Infrastructure Report 2024 Conference with IDFC Foundation

National Conference on Sustainable Cooling
(In collaboration with BEE & Ministry of Power)

Participation in **14th Clean Energy Ministerial and 8th Mission Innovation meeting (CEM14/MI-8)**

Distributed Renewable Energy for Sustainable Development with Climate Policy Initiative

Upcoming Dissemination Workshops

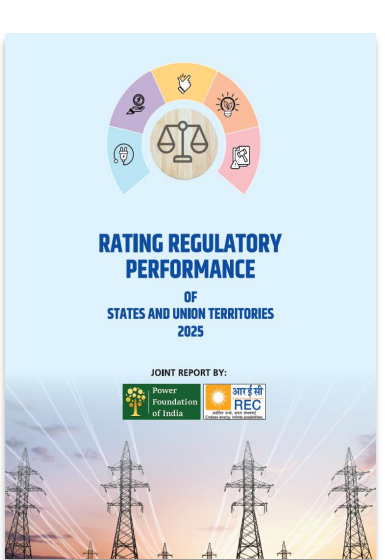
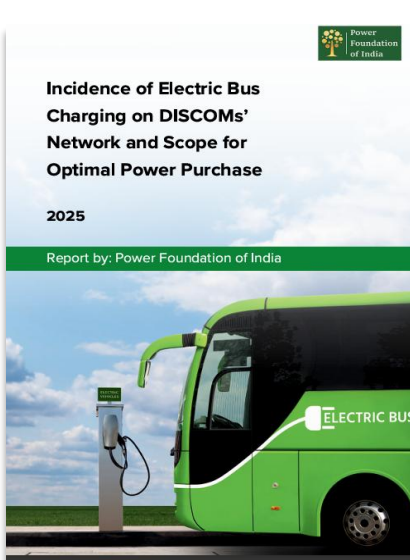
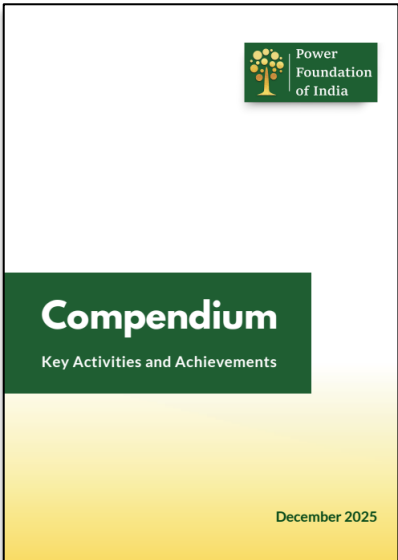
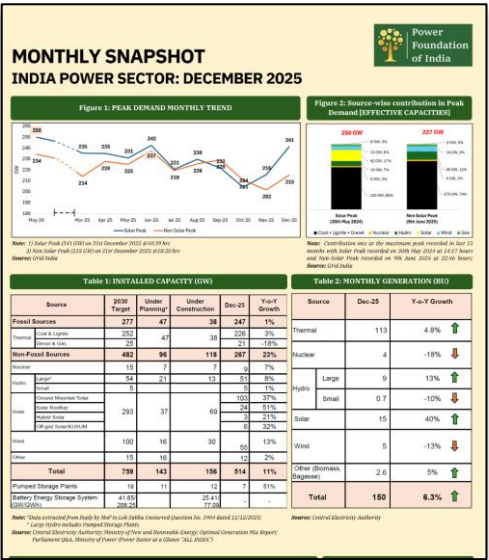
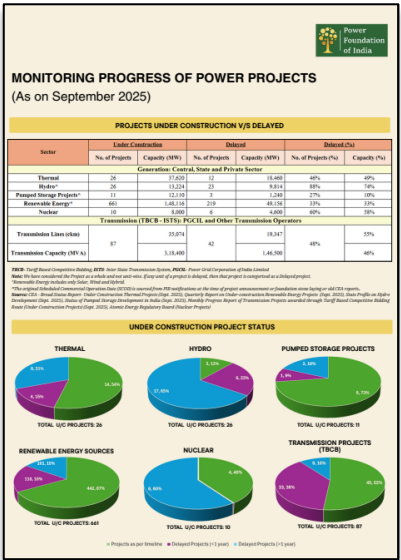
- *Tapping the Potential of Agriphotovoltaics in India* – March 2026
- *Strategic Pathways for Energy Storage in India through 2032* – April 2026

Proposed Campaign: Adoption of Prepaid Smart Meter (to be reviewed)

Communication and Outreach

Creative Content & Multimedia Production

This vertical/function delivers end-to-end multimedia content designing and development. From concept ideation to execution, the team creates impactful communication assets across multiple formats - Reports, Compendium, Videos, Newsletters and other digital and print collaterals.



- Digital & Social Media**

Structured social media outreach across platforms, with **LinkedIn** emerging as the highest-engagement platform, with over **9,000 organic followers** and an average growth of **150+ new followers per month**.

- Website Update and Management**

Communication and Outreach

SAMARTH – National Biomass Mission

Awareness Campaign

- **Objective:** Highlight the adverse impact of burning Parali (paddy residue) and encourage pellet manufacturing.
- **Formats:** Print advertisements & outdoor campaigns (hoardings, OOH)
- **Geographic Coverage:** Delhi NCR, Haryana, Punjab, Western Uttar Pradesh, Rajasthan
- **Languages:** English, Hindi, Punjabi



(Print Ads)

Skill Development Program

- In association with Power Sector Skill Council
- Period: March 2025 – Ongoing
- Objective: Training of pellet plant workers on health & hygiene, safety protocols, and plant operations
- 1,500 pellet plant workers to be trained by Nov 2027. 550 workers trained across 23 pellet plants
- 6 States – Uttar Pradesh, Haryana, Punjab, Rajasthan, Madhya Pradesh, and Delhi NCR

Impact Assessment Study

- To assess outcomes and on-ground impact of National Biomass Mission interventions during the last five years.

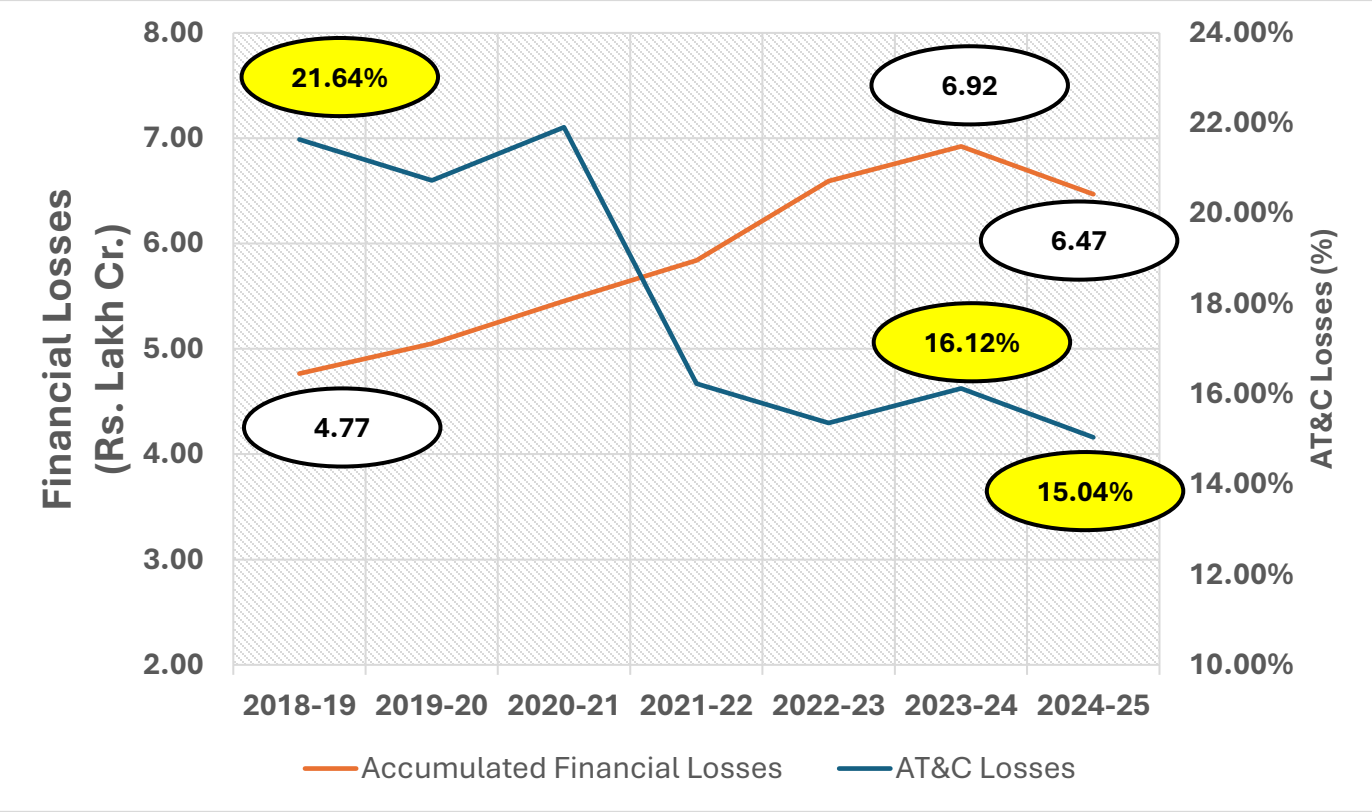
India Smart Utility Week 2026

Smart Energy & Smart Mobility

DISCUSSION POINTS



Operational improvements but financial losses keep mounting



Source: PFC Performance of Power Utilities Reports

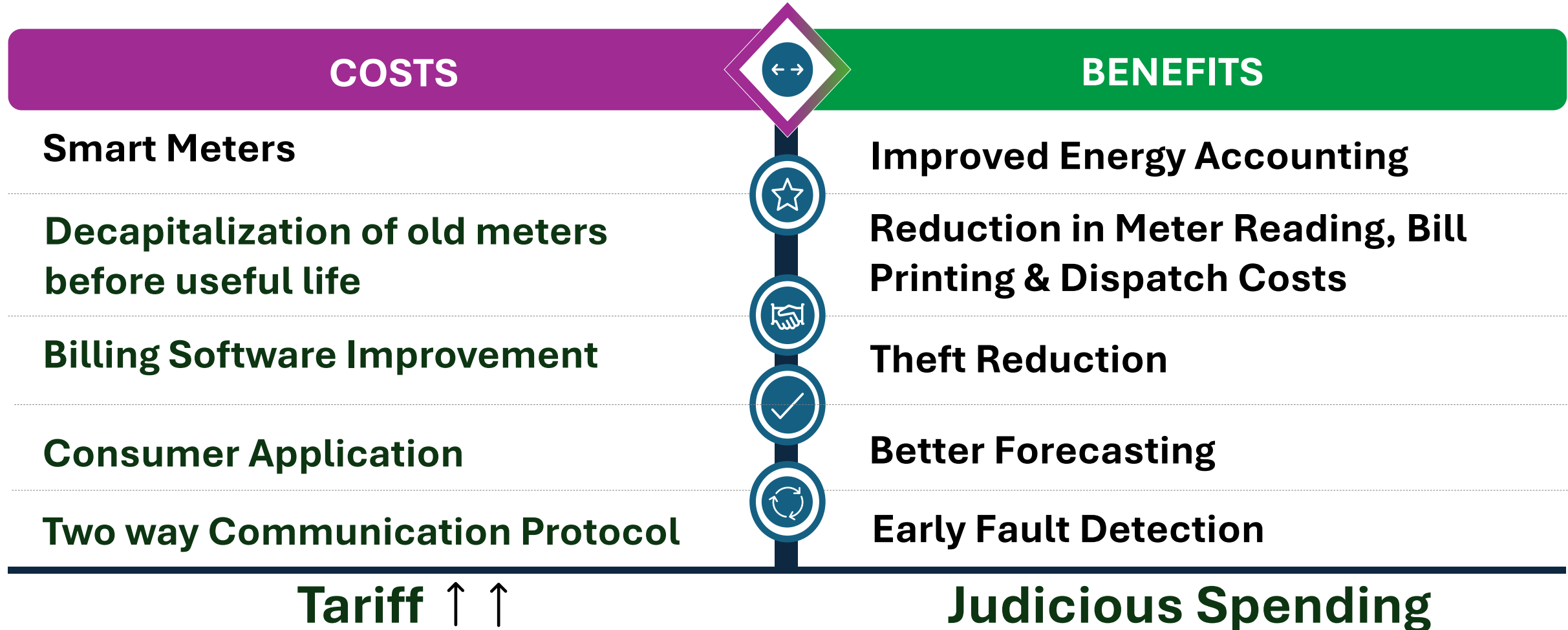
- **Net worth** of DISCOMs has worsened from **Rs. (-) 63,588 Cr. in FY 2018-19** to **Rs. (-) 1,44,711 Cr. in FY 2023-24**

Distribution Reform Schemes

Scheme	Outlay (Rs. Cr.)
Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY)	33,000
Accelerated Power Development and Reforms Programme (APDRP)	40,000
Restructured APDRP (R-APDRP)	51,577
Integrated Power Development Scheme (IPDS)	32,612
Deen Dayal Upadhyay Gram Jyoti Yojana (DDUGJY)	75,893
Pradhan Mantri Sahaj Har Ghar Bijli Yojana (Saubhagya)	16,320
Ujjwal DISCOM Assurance Yojana (UDAY)	State Govt. to take over 75% DISCOM Debt
Revamped Distribution Sector Scheme (RDSS)	3,03,758

SMART GRID will come at a cost

$$\begin{array}{l}
 \text{SMART GRID} \\
 \text{(DIGITAL GRID)}
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 \text{SMART CONSUMER} \\
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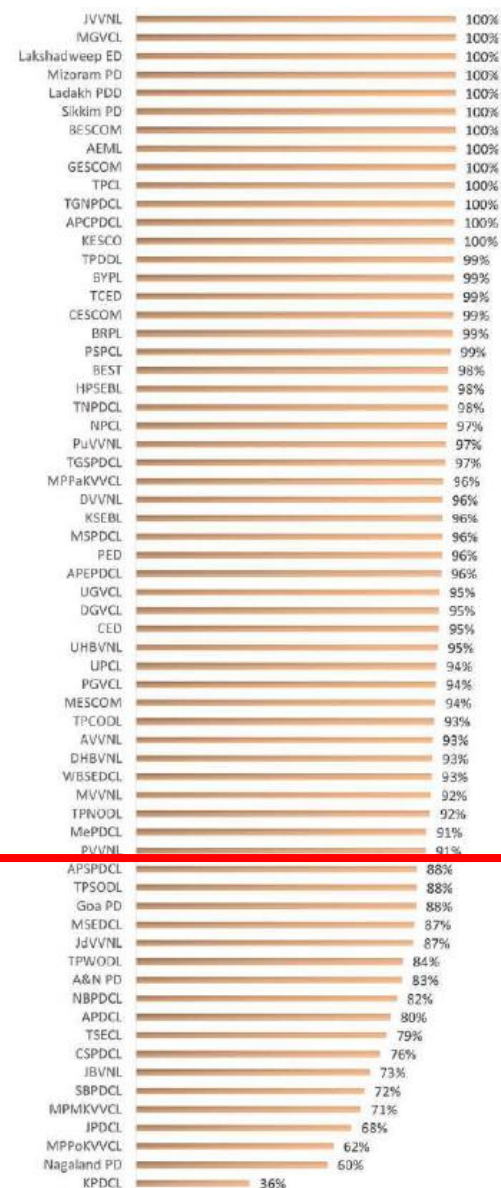


Is India ready for Smart DISCOM?

Particulars	Unmetered Nos. 2025	% Unmetered 2023	% Unmetered 2025	Smart Meters Sanctioned under RDSS	Smart Meters Communicating
Feeder (66 kV/ 33 kV/ 22 kV)	1,653	1.38%	3.85%	1.9 L	1.5 L
Feeder (11 kV)	3,990	0.77%	1.54%		
DT	88,62,451	64.1%	57.70%	52 L	14 L
Consumers	20,621,149	5.32%	5.62%	1,972 L	427 L

Source: CEA Status of Metering 2023 & 2025, Smart Meter Portal

2.4.2 Bills Generated Based On Actual/ Working Meter Reading



Actual metered reading ensures consumers are billed accurately based on their precise electricity consumption, promoting transparency and fairness.



FAST FACTS

National Maximum:
100%

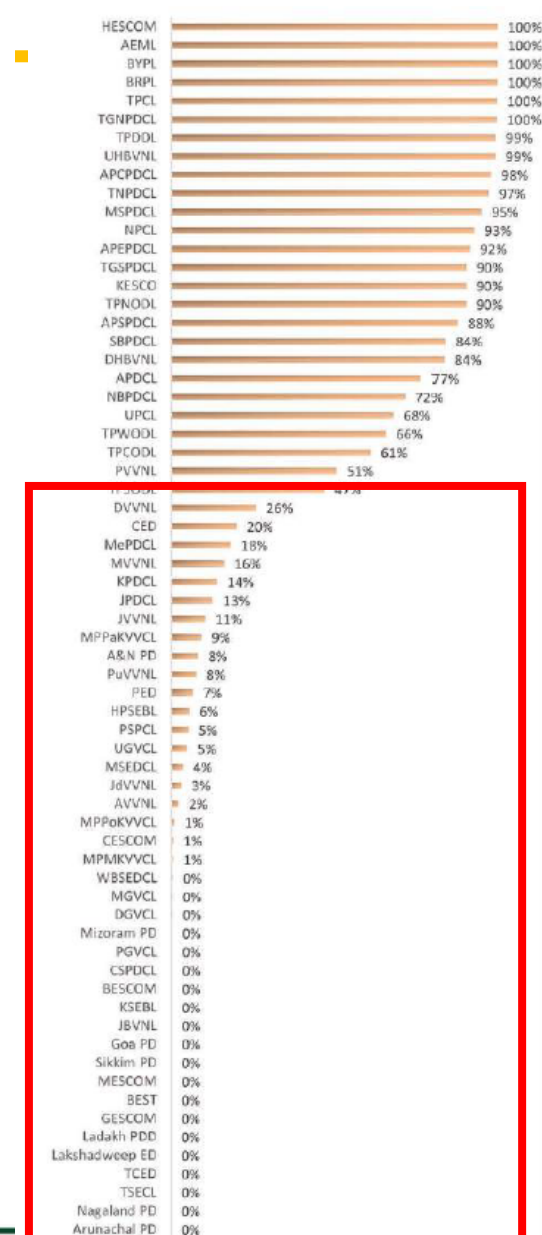
National Average:
91%

National Minimum:
36%

TAKEAWAYS

- Three DISCOMs—MPPoKVVCL, Nagaland PD and KPDCL generated less than 65% of bills through actual meter readings, indicating a need for them to reassess their practices in maintaining the functionality of the electricity meters
- 47 DISCOMs exceed the national average in bill generation based on actual meter reading
- 7 DISCOMs—LPDD, Sikkim PD, Mizoram PD, Lakshadweep ED, MSPDCL, UPCL, and MePDCL generate a higher proportion of bills based on actual meter readings than the national average

2.4.3 Bills generated based on non-manual Meter Reading



Non-manual meter readings, a subset of actual meter readings, indicate the meter readings captured via all automated means other than the traditional manual reading mode which is with bare eyes.



FAST FACTS

National Maximum:
100%

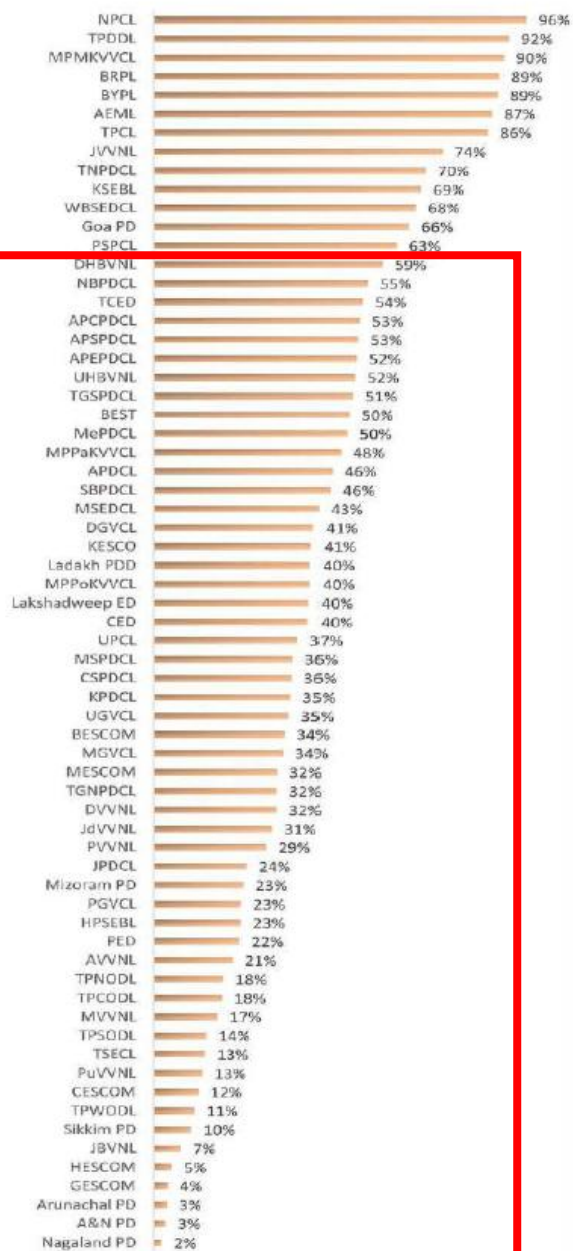
National Average:
37%

National Minimum:
0%

TAKEAWAYS

- The National Average has increased by 7% compared to the previous year, primarily due to the incremental installation of smart meters on consumer premises
- Few DISCOMs demonstrate excellence with a commendable 100% of bill generation based on automated meter readings: AEML, HESCOM, BYPL, BRPL, TPCL, and TGNPDCL
- In contrast, Ladakh PDD, Lakshadweep ED, TCED, TSECL, and Nagaland PD currently have a 0% reliance on automated meter reading methods, presenting an opportunity for future advancements in this area

2.4.9 % Bills Paid Digitally



Promoting digital payments supports broader governmental initiatives to digitize services and increase technological integration in utilities. Additionally, benefits consumers by offering convenience, faster transactions and secure payments.



FAST FACTS

National Maximum:
96%

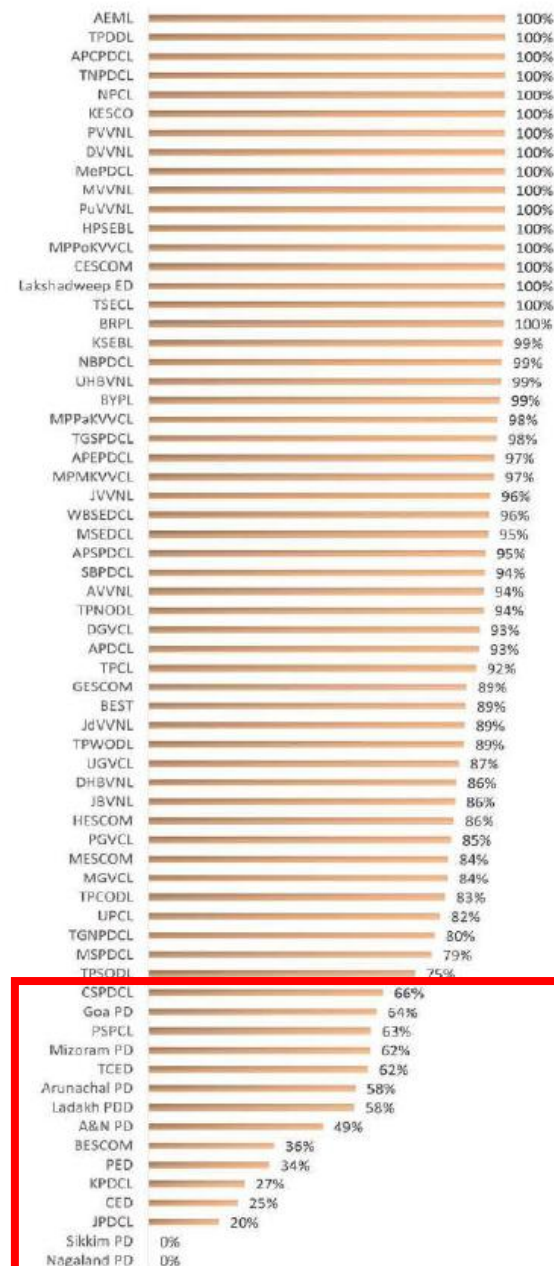
National Average:
41%

National Minimum:
2%

TAKEAWAYS

- The leading DISCOMs, where over 80% of bills are paid via online modes, include 7 DISCOMs: AEML, NPCL, MPMKVVCL, BRPL, TPDDL, BYPL, and TPCL.
- Additionally, there are 29 DISCOMs across 17 states and Union Territories where the proportion of consumers paying digitally exceeds the national average of 41%.

2.4.6 Percentage of Consumers Receiving Billing Alerts



Sending timely billing alerts to consumers gives them visibility into their electricity bills and consumption, allowing them adequate time for making the payments to the DISCOM and avoid late payment fee etc.



FAST FACTS

National Maximum:
100%

National Average:
82%

National Minimum:
0%

TAKEAWAYS

- The foremost DISCOMs providing billing alerts to consumers at a rate of 100% include 17 DISCOMs across 9 states and Union Territories: TNPDC, AEML, NPCL, KESCO, MPPoKVVCL, PVVNL, DVVNL, MVVNL, PuVVNL, CESCOM, APCPDCL, HPSEBL, TSECL, TPDDL, MePDCL, Lakshadweep ED, and BRPL.
- Furthermore, there are 48 DISCOMs that facilitate billing alerts to consumers at a rate exceeding the national average of 82%.

Billing Errors Still Prevalent

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Labourer receives electricity bill for ₹1.61 crore

Published: September 04, 2025 07:53 pm IST - TIRUNELVELI



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Today's ePaper

'Heart attack moment': UP farmer stunned by Rs 7.03 crore electricity bill

TNN / Updated: Feb 9, 2025,

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News / Viral News / ₹222 Crore Electricity Bill Sent To Noida Farmer In Shocking Blunder

Updated 30 December 2025 at 20:57 IST

₹222 Crore Electricity Bill Sent To Noida Farmer In Shocking Blunder

The electricity department handed over an electricity bill amounting to Rs 222 crore to a farmer in Noida. The bill sent to him showed arrears for seven months, even though the farmer claimed that he paid bills every month.

Feb 24, 2026 2:38 PM

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Tamil Nadu

A missing dot turns Rs 494 power bill into Rs 1.61 crore shocker for Nellai family

Improper Energy Accounting

- Huge Unmetered Sales persist
- DISCOMs are also claiming **new unmetered connections** -> Against MoP (RoC) Rules, 2020
- Results in **improper Energy Accounting**
- Energy Balance based on estimations & hence Distribution Losses do not reflect actual figures

State	Unmetered Sales (MU)	% of Total Sales
Tamil Nadu	15,909	17%
Punjab	12,681	21%
Madhya Pradesh	28,149	39%
Maharashtra	39,561	30%
Total	96,300	27%

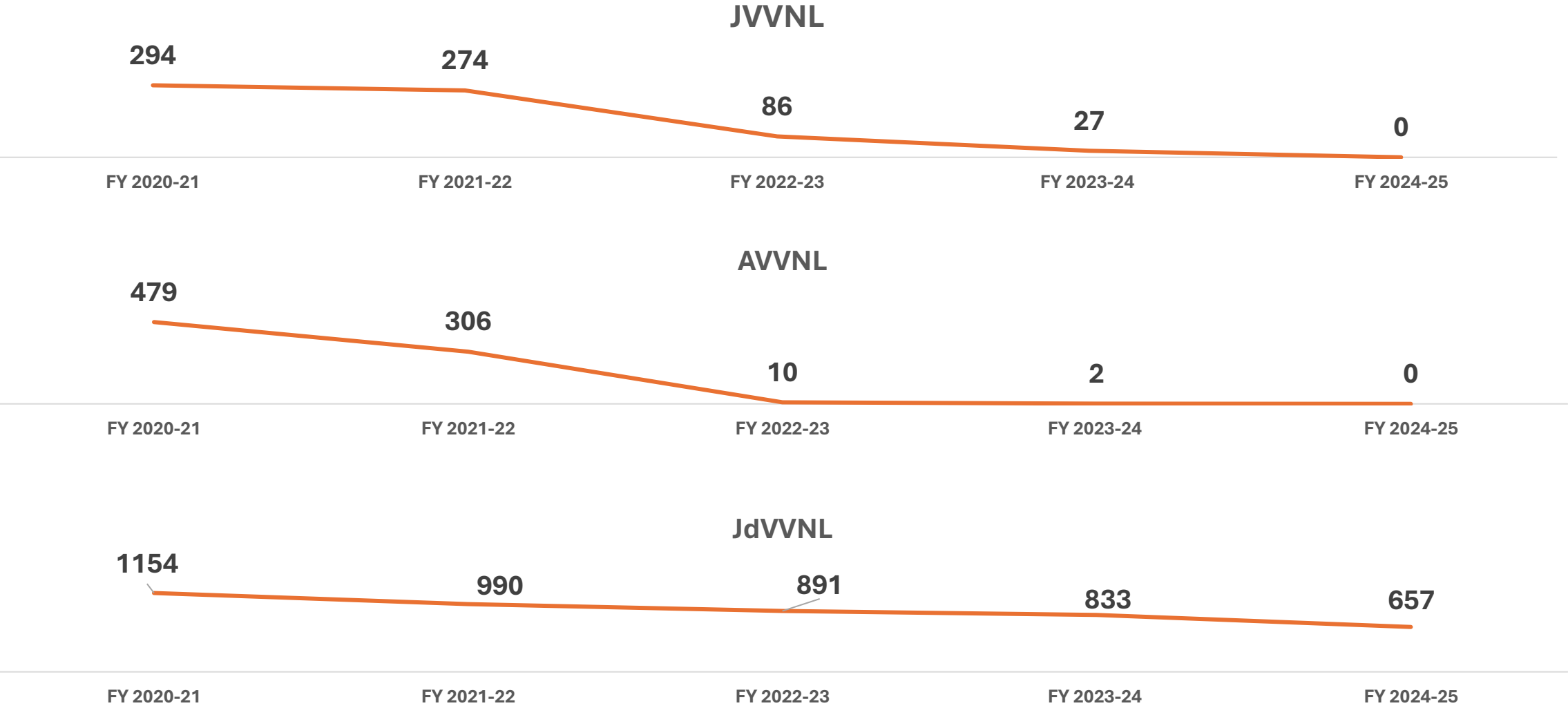
Source: FY 2023-24 True-Up Petitions

Illustration: Unmetered Sales can Mask Inefficiency

Scenario	Metered Sales (MU)	Unmetered Sales (MU)	Distribution Losses (MU)	Power Purchase Quantum (MU)
Scenario 1	100	30	10	140
Scenario 2	100	25	15	140

Improvement witnessed in few DISCOMs - RAJASTHAN

Unmetered Sales (MU)



Increasingly Sophisticated Forecasting & Scheduling Required (1/2)

MP DISCOMs Tariff Petition FY 2026-27

Table 81: Projection of Hourly Scheduling of Solar Energy for FY 2026-27 (MUs)

Hourly Schedule Summary (MUs)													
Existin g ToD	Dates /Hour s	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27
Off Peak Night	00-01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	01-02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	02-03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	03-04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	04-05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mornin g Peak	05-06	1.94	1.99	1.89	1.97	1.93	2.00	2.21	2.19	2.28	2.28	2.59	2.79
	06-07	25.34	27.98	22.79	23.74	17.37	18.12	13.41	9.13	7.37	7.39	24.17	33.70
	07-08	64.48	66.13	63.00	57.51	56.46	58.80	57.58	48.93	54.53	54.91	79.54	85.75
	08-09	97.83	104.47	91.86	85.53	84.02	87.32	90.87	94.97	105.65	109.03	132.78	142.67
Off Peak/ Solar Hours	09-10	123.40	120.59	115.08	117.72	113.56	115.94	124.19	129.24	149.10	152.51	169.42	176.57
	10-11	142.99	142.73	128.40	127.51	133.42	136.62	148.66	149.76	173.88	184.62	201.62	210.19
	11-12	158.77	162.95	147.78	147.64	151.33	157.38	166.42	170.16	195.89	204.26	228.37	241.27
	12-13	158.75	162.84	149.69	151.66	153.27	155.21	168.67	172.48	198.46	204.49	231.15	244.11
	13-14	158.75	158.72	145.77	149.57	143.21	148.97	159.75	163.49	193.66	202.11	225.90	235.66
	14-15	140.98	138.39	128.24	133.38	131.30	136.56	141.93	147.59	169.26	182.45	212.38	218.69
	15-16	113.41	110.19	101.17	107.14	115.41	107.55	106.34	106.56	127.28	142.90	163.89	173.38
	16-17	84.03	84.07	78.25	81.30	77.95	76.94	64.09	67.47	77.90	88.67	113.53	122.79
Evening Peak	17-18	40.94	43.92	41.82	43.41	42.65	34.09	26.45	28.07	31.52	31.71	57.09	60.12
	18-19	9.73	11.95	20.93	19.67	13.46	7.95	6.62	4.56	5.03	5.05	5.68	5.73
	19-20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	20-21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	21-22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Off Peak Night	22-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	23-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 82: Projection of Hourly Scheduling of Wind Energy for FY 2026-27 (MUs)

Hourly Schedule Summary (MUs)													
Existin g ToD	Dates /Hour s	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27
Off Peak Night	00-01	19.77	26.11	26.27	25.97	23.07	19.43	18.69	14.18	13.03	13.02	13.33	16.83
	01-02	21.36	27.69	28.53	28.32	24.57	20.87	19.54	15.02	13.90	13.89	14.12	18.51
	02-03	23.73	30.13	30.53	29.90	25.99	22.78	21.24	16.69	15.64	15.63	15.69	20.19
	03-04	25.32	31.76	32.05	32.30	28.19	24.26	22.94	18.35	17.38	17.36	17.25	21.88
	04-05	26.91	33.40	34.36	33.93	29.83	25.82	23.79	20.02	19.11	19.10	18.82	23.56
	05-06	28.52	35.10	36.41	35.63	32.25	28.06	25.49	21.66	20.82	21.68	20.36	25.25
Mornin g Peak	06-07	30.20	36.84	37.31	37.38	33.16	29.04	27.19	24.49	24.29	26.17	22.84	27.06
	07-08	28.69	35.31	36.69	35.88	32.59	28.48	25.49	23.29	23.93	24.79	22.32	25.68
	08-09	27.15	33.76	35.24	34.34	31.12	26.97	23.79	21.35	21.76	21.67	20.43	23.95
Off Peak/ Solar Hours	09-10	23.98	30.48	31.38	31.15	27.99	23.93	21.24	17.72	17.43	17.36	16.72	20.52
	10-11	21.58	28.02	29.79	29.47	26.45	22.44	19.54	15.91	15.44	15.36	14.95	18.77
	11-12	20.00	26.38	27.48	27.05	24.89	20.94	18.69	14.99	14.43	14.35	14.08	17.06
	12-13	17.60	23.89	25.13	24.59	22.48	18.61	16.14	12.35	11.54	11.49	11.59	14.50
	13-14	16.00	22.23	23.55	22.93	20.85	17.04	14.44	10.60	9.63	9.59	9.95	12.79
	14-15	17.58	23.83	25.06	24.53	22.42	18.56	16.14	12.39	11.61	11.54	11.62	14.50
	15-16	19.94	26.28	27.33	26.88	24.77	20.81	18.69	15.12	14.64	14.54	14.21	17.06
	16-17	21.52	27.89	29.67	29.31	26.28	22.29	19.54	16.09	15.96	15.86	15.40	18.82
Evening Peak	17-18	23.84	30.26	31.09	30.79	27.72	23.67	21.24	17.99	18.18	18.03	17.31	20.55
	18-19	25.38	31.80	32.57	32.33	29.16	25.13	22.94	19.19	19.29	19.14	18.87	22.29
	19-20	26.95	33.32	33.82	33.87	30.49	26.72	23.79	20.13	19.26	19.25	18.97	23.65
	20-21	28.53	34.96	35.35	35.49	31.37	27.38	25.49	21.69	20.83	21.68	20.36	25.25
	21-22	30.08	36.58	36.02	36.95	31.99	28.00	27.19	23.36	22.57	24.30	21.95	26.94
Off Peak Night	22-23	28.49	35.00	35.26	35.33	31.29	27.27	25.49	21.69	20.84	21.70	20.38	25.25
	23-24	26.91	33.36	33.73	33.72	29.75	25.87	23.79	20.02	19.10	19.10	18.82	23.56

Increasingly Sophisticated Forecasting & Scheduling Required (2/2)

MP DISCOMs Tariff Petition FY 2026-27

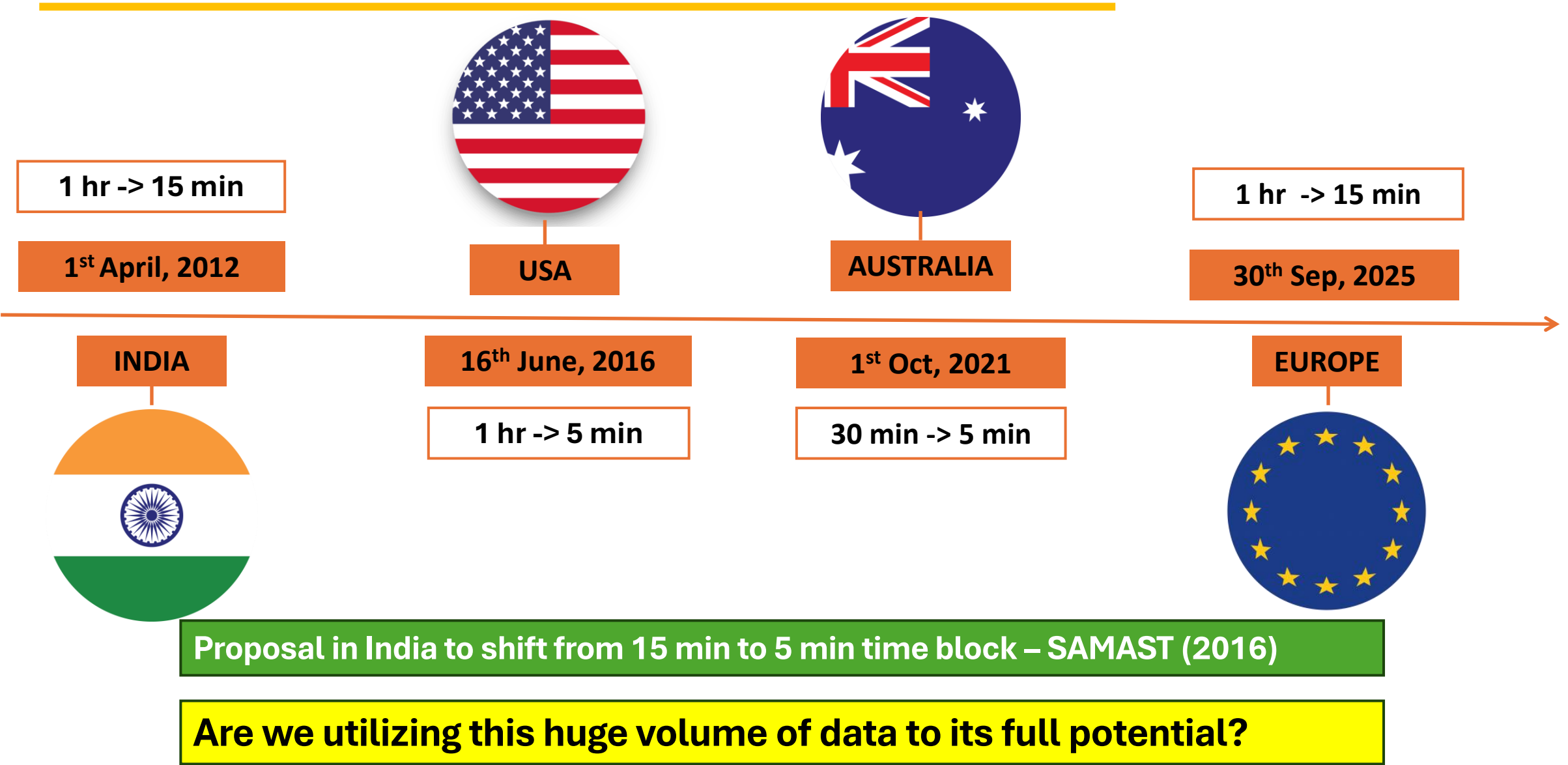
Table 83: Projection of Hourly Scheduling of Other Must Run Energy for FY 2026-27 (MUs)

Hourly Schedule Summary (MUs)													
Existin g ToD	Dates /Hour s	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27
Off Peak Night	00-01	32.97	35.51	32.24	42.35	73.99	72.48	53.82	38.70	42.18	50.72	37.17	36.44
	01-02	31.57	34.41	31.96	40.71	73.48	72.21	53.21	38.08	41.43	50.35	37.11	35.98
	02-03	30.01	33.56	32.06	40.69	66.04	66.18	52.52	37.64	33.81	41.87	36.85	35.28
	03-04	29.92	33.24	31.79	41.70	67.72	66.08	52.46	37.63	35.63	41.95	36.96	35.17
	04-05	30.17	33.39	31.76	41.92	79.01	77.00	52.64	38.57	41.40	42.10	37.36	35.43
	05-06	36.23	38.28	36.66	42.62	81.79	79.93	69.57	66.18	64.44	63.47	57.70	48.57
Mornin g Peak	06-07	44.84	40.13	37.61	44.03	82.79	82.81	75.92	71.31	86.51	84.57	69.18	60.65
	07-08	44.30	40.12	37.84	43.76	83.49	84.28	73.41	74.97	95.52	93.91	76.54	64.11
	08-09	37.90	39.48	37.47	43.86	83.21	83.79	72.17	74.77	86.93	86.62	69.61	57.11
Off Peak/ Solar Hours	09-10	36.73	38.50	36.95	43.55	71.59	71.31	70.43	73.81	79.34	80.55	66.36	55.01
	10-11	20.83	21.06	18.84	19.62	71.17	70.20	53.31	31.39	33.70	37.65	29.24	22.18
	11-12	21.24	21.41	18.78	19.23	71.43	70.09	52.74	29.32	31.64	35.29	27.67	21.21
	12-13	20.98	21.05	19.03	19.14	71.29	69.51	52.14	28.05	30.85	32.65	26.02	21.24
	13-14	21.00	21.02	19.26	19.04	71.17	69.65	52.46	28.17	30.51	31.68	25.71	20.14
	14-15	20.93	20.95	19.07	19.24	71.27	69.79	52.86	28.92	30.64	31.19	25.49	20.70
	15-16	20.86	34.50	18.51	19.18	71.20	70.73	54.40	31.33	33.51	32.60	26.58	21.61
	16-17	38.10	44.12	42.67	42.28	82.77	82.15	71.28	75.02	88.78	86.24	66.18	50.03
Evening Peak	17-18	39.73	43.69	42.76	43.07	82.77	83.11	72.44	77.23	91.94	87.71	68.30	50.23
	18-19	44.22	44.66	43.50	44.07	83.88	84.10	72.98	73.05	92.64	89.81	73.27	55.31
	19-20	45.95	46.29	43.06	44.74	83.93	90.28	76.18	67.83	82.51	80.65	66.75	55.52
	20-21	42.89	46.33	43.78	45.06	70.42	72.40	50.23	35.74	53.64	60.86	59.01	50.13
Off Peak Night	21-22	32.03	35.87	33.25	44.67	69.26	71.11	53.61	27.55	46.13	42.84	37.70	36.43
	22-23	32.52	36.80	33.20	44.03	76.41	70.69	54.56	38.54	46.44	42.42	37.84	36.60
	23-24	32.78	36.56	32.66	43.54	76.34	80.50	54.68	39.23	46.51	42.37	37.30	36.40

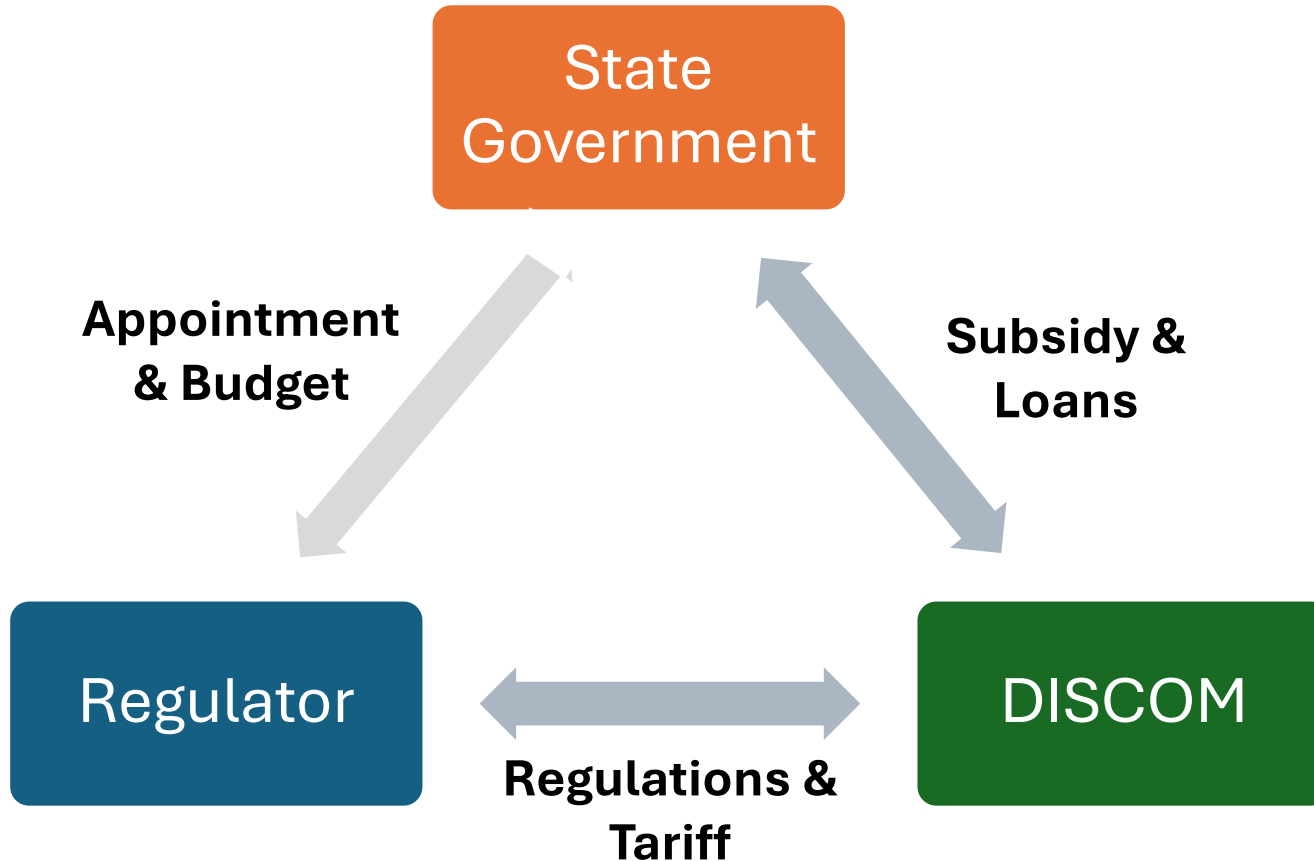
Table 84: Projection of Hourly Scheduling of Thermal Energy (TMM+MOD) for FY 2026-27 (MUs)

Hourly Schedule Summary (MUs)													
Existin g ToD	Dates /Hour s	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27
Off Peak Night	00-01	279.66	296.45	271.73	239.92	201.13	205.63	249.24	288.93	265.55	260.72	288.27	292.02
	01-02	274.71	289.82	262.89	235.27	196.11	200.77	243.84	275.32	253.26	253.21	279.91	281.07
	02-03	264.88	280.03	254.55	229.33	197.62	200.48	236.73	268.97	250.88	253.82	273.00	268.99
	03-04	259.81	273.56	248.29	223.31	191.35	197.55	234.41	272.58	252.82	256.16	275.42	268.77
	04-05	254.74	267.45	242.22	219.94	177.97	184.69	234.60	273.35	250.17	254.04	272.02	271.36
	05-06	249.30	262.77	239.26	224.48	180.56	186.98	226.51	252.47	240.53	238.69	250.29	262.73
Mornin g Peak	06-07	237.98	242.22	226.46	221.17	191.12	194.45	242.69	293.69	276.40	267.66	259.64	261.26
	07-08	210.32	203.80	183.16	192.43	162.36	165.92	222.95	304.59	300.32	297.65	250.83	233.69
	08-09	177.68	158.00	144.67	158.71	133.01	133.21	188.03	282.09	294.85	291.08	228.31	185.34
Off Peak/ Solar Hours	09-10	151.63	143.40	121.03	121.99	108.74	108.32	149.77	262.36	279.84	277.75	210.80	150.85
	10-11	146.81	145.69	129.02	128.47	79.95	79.71	134.52	282.30	297.90	299.26	226.47	139.28
	11-12	133.38	140.56	122.60	107.73	56.44	55.80	115.05	254.49	263.54	274.90	194.81	88.74
	12-13	134.08	149.46	129.77	101.21	47.89	52.36	106.89	245.85	251.59	265.18	181.96	65.87
	13-14	136.39	159.68	139.54	102.30	54.95	59.03	117.33	259.54	258.02	267.51	187.09	68.24
	14-15	156.53	182.26	158.68	118.79	65.09	69.90	135.87	268.50	270.10	274.76	191.28	77.51
	15-16	184.78	195.94	181.16	144.44	79.85	94.74	167.14	303.16	306.61	306.99	238.41	125.98
	16-17	185.12	191.91	163.15	142.18	103.45	109.99	187.59	289.73	295.09	296.53	234.47	149.18
Evening Peak	17-18	207.90	201.96	180.40	179.50	143.36	156.02	229.49	311.42	323.69	332.22	266.18	212.09
	18-19	231.90	223.42	196.05	206.51	183.83	200.00	267.00	308.20	313.87	329.02	286.72	261.08
Off Peak Night	19-20	257.59	256.53	237.65	239.74	213.68	209.84	258.29	281.05	293.80	312.33	280.12	274.95
	20-21	247.04	252.56	239.97	232.52	211.00	206.77	252.83	265.47	269.72	284.89	245.65	249.82
	21-22	252.91	261.32	248.17	224.55	202.15	199.14	237.34	256.84	243.54	266.37	247.09	249.14
Off Peak Night	22-23	273.26	283.25	259.95	232.55	197.02	203.03	243.81	276.66	258.21	276.91	277.48	279.87
	23-24	280.51	292.47	265.09	235.70	196.44	193.93	246.08	287.78	260.85	274.89	288.75	290.06

Developed World has only recently evolved



Interplay of key bodies needs to be efficient



State Government

- Timely Payment of Subsidy

Regulator

- Normative parameters in touch with ground reality
- Timely issuance of Tariff Order
- Cost-reflective Tariff

DISCOM

- Meet normative parameters
- Timely filing of Tariff Petition
- Strategic Power Procurement
- Increase operational efficiency
- Levy monthly FPPAS as per Regulations

India Smart Utility Week 2026

REGULATIONS FOR DIGITAL GRIDS & PEER TO PEER TRANSACTIONS



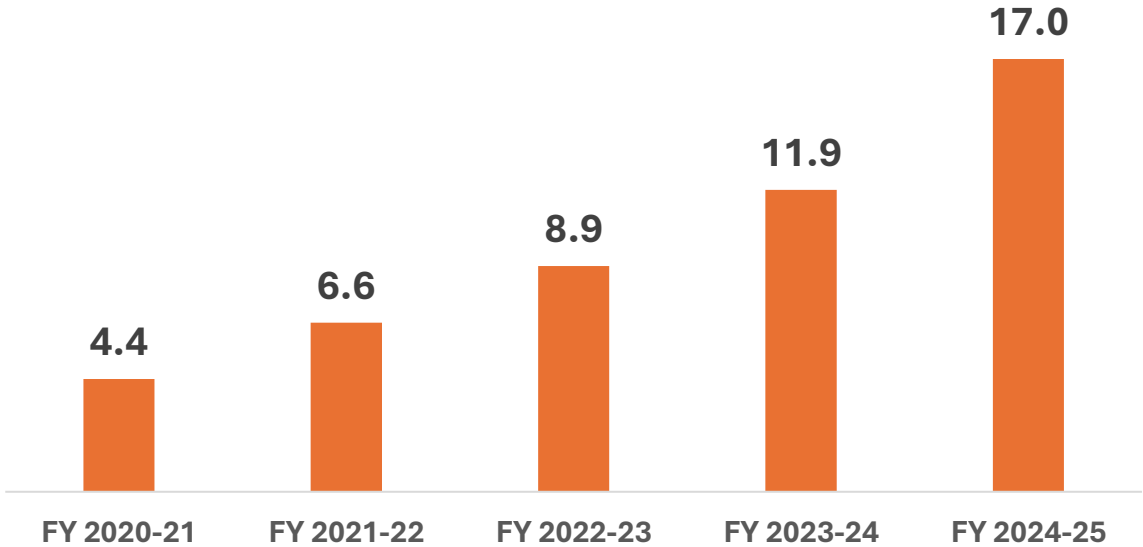
Need: P2P Energy Transactions

- **PM Surya Ghar: 1 Cr. Households** by FY 2026-27
- MoP has Notified Separate Distributed Renewable Energy Trajectory

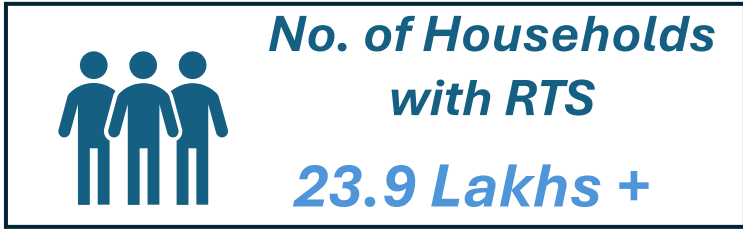
Year	DRE Target
2024-25	1.50%
2025-26	2.10%
2026-27	2.70%
2027-28	3.30%
2028-29	3.90%
2029-30	4.50%

- **Rooftop Solar Faster Adoption – need of the hour**
- **Optimization of PPC**

Rooftop Solar Installed Capacity (GW)

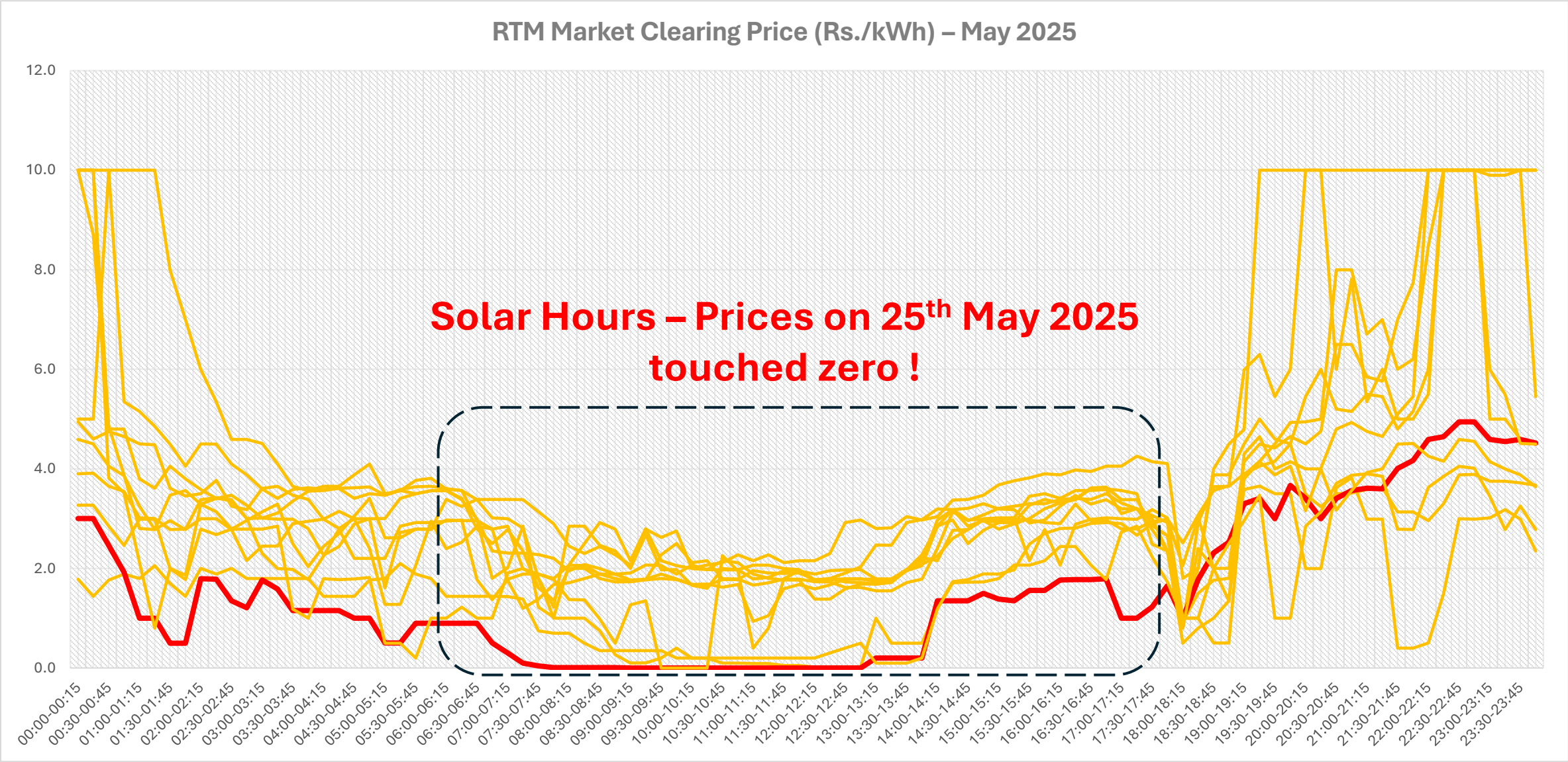


Source: MNRE State wise Cumulative Installed Capacity



Source: PIB, India's Solar Momentum

Solar Hours – extremely low prices



Source: IEX 20th-30th May 2025 RTM MCP

Innovations in Net Metering - VNM & GNM

Net Metering	Virtual Net Metering	Group Net Metering
Injection and drawl point same—i.e., the same premise or building.	Injection point and drawl point are different. Solar panels at one location; virtually allocate the generated power to one or multiple electricity connections	Injection point and drawl point are different. Single consumer having multiple service connections across various locations



Physical Space Constraint Addressed

Need: P2P Energy Transactions

- Under Prevailing Net-Metering Regulations, **treatment of surplus energy at the end of settlement period varies**
- Prosumers in Type 2 States – No compensation for energy surplus injected

Type 1	Type 2
Surplus Settled at APPC	Surplus Lapses
Delhi	Bihar
Kerala	Haryana
Assam	Rajasthan



Lower Transmission Losses



DISCOM benefits in case of subsidized consumers



Consumers get Green Power at a lower Cost



Community Grid

P2P Energy Transactions – Better Consumer Incentivization Model

Legal Status: P2P Energy Transactions

- Peer-to-Peer Transactions **can not be considered as Trading**
- **Pilots** have been conducted in **Regulatory Sandbox Environment**
- Regulations / Guidelines issued till date use the term “**Peer-to-Peer Transaction**”
- Service Provider (P2P Platform) **is not a licensed entity**

- “2(71) ***"trading" means purchase of electricity for resale thereof and the expression "trade" shall be construed accordingly;***”
- “Section 12 (Authorized persons to transmit, supply, etc., electricity) mandates that
“No person shall
(a) *transmit electricity; or*
(b) *distribute electricity; or*
(c) undertake trading in electricity,
unless he is authorized to do so by a license issued under section 14, or is exempt under section 13.”

P2P Transactions involve an additional Virtual Layer

Physical Layer includes infrastructure like RE sources (solar PV, wind turbines, etc.), energy storage systems (batteries), and smart meters to support electricity generation, storage, and distribution in a Community Grid.

Virtual Layer is a combination of digital platforms/ software systems to utilize technologies like blockchain and smart contracts enabling secure, transparent energy transaction and real-time data analytics, for grid operations.

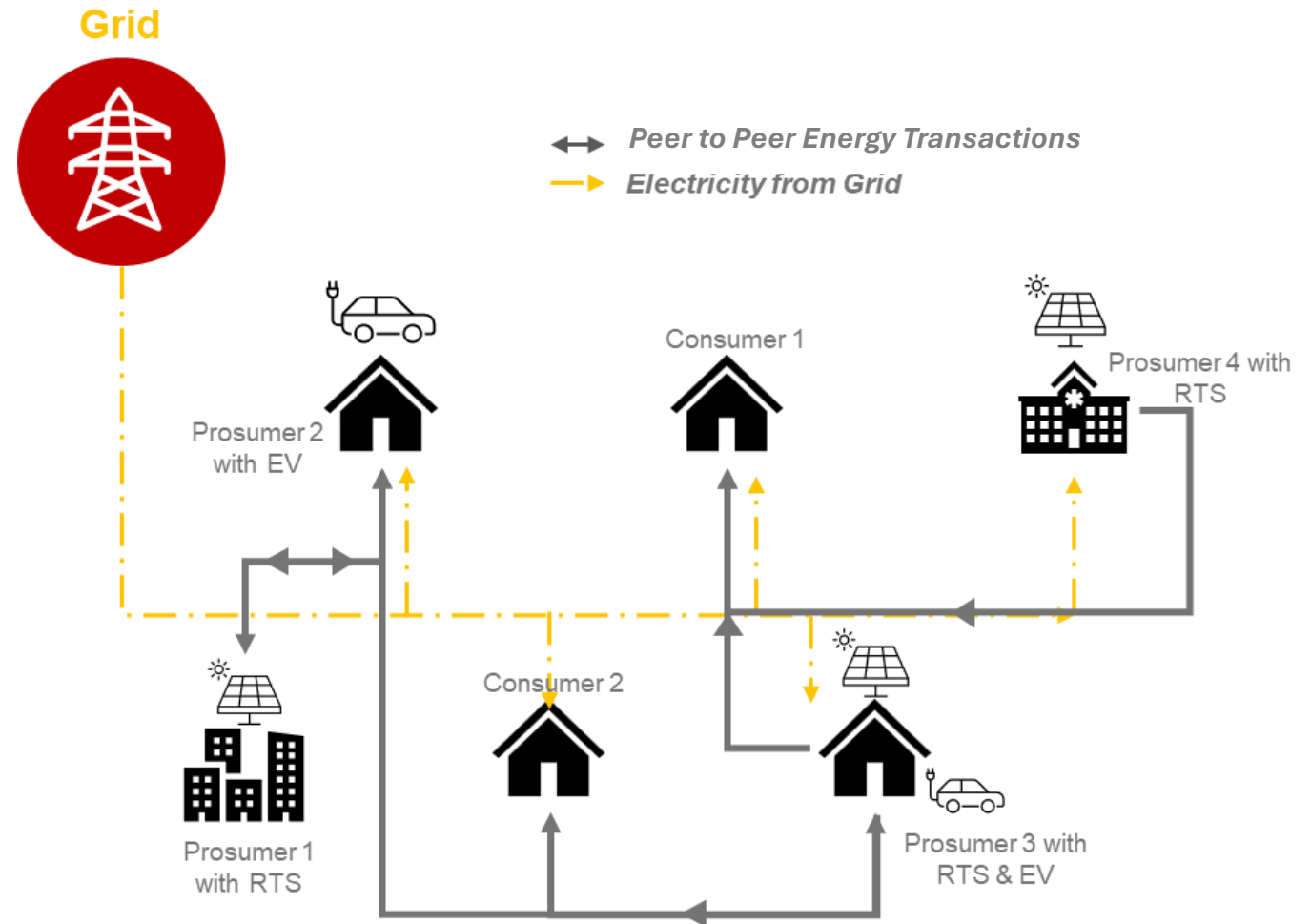


Figure 1: Peer-to-Peer Transaction Flow in a Community Grid

P2P Regulations / Guidelines Notified in 4 States

Particulars	Uttar Pradesh (UPERC)	Delhi (DERC)	Karnataka (KERC)	Kerala (KSERC)
Regulations/ Guidelines	Guidelines for peer-to-peer solar energy transaction through blockchain based platform, 2023	DERC (Peer to Peer Energy Transaction) Guidelines, 2024	KERC (Implementation of Peer to Peer Solar Energy Transaction) Regulations 2024	KSERC (Renewable Energy & Related Matters) Regulations, 2025
Eligibility/ Requirements	- Post-paid smart meter, at own cost - Prosumers with solar rooftop can only be seller - Net-Metering facility for Prosumers	- ToD compliant energy meter/ smart meter - Less than 200 kW sanctioned load Capacity of DRE capped at 5x of sanction load Can use only one: Virtual/ Group/ P2P net-metering	- ToD compliant energy meter/ smart meter - Domestic consumers or SRTPV prosumers	- Prosumers with DRE systems - Only surplus energy can be sold
Ledger Technology	Blockchain	Blockchain / any other technology	Blockchain / any other technology	Blockchain / any other technology
Transaction Price	Agreed on P2P Platform	Agreed on P2P Platform	Preferential: mutually negotiated rate - Dynamic: cleared price as per P2P platform	To be decided after conducting a Pilot
Pricing/ Charges	- P2P transaction charge: Rs. 0.41/ kWh - Open Access charges: Wheeling + CSS, AS Waiver	- P2P transaction charge: determined by DERC Open Access charges: Nil	- P2P transaction charge: Rs. 0.14/kWh - Green Open Access charges; Nil if on same DT	To be decided after conducting a Pilot
Scheduling/ Billing	- Submit Day-Ahead schedules or 4 time-blocks before for intra-day - Billing as per DISCOM cycle	- Submit schedules 8 time blocks prior - Billing as per DISCOM cycle	- Submit Day-Ahead schedules or 4 time-blocks before for intra-day - Billing as per DISCOM cycle	To be decided after conducting a Pilot

India Smart Utility Week 2026

Regulation for Digital Grids

CHALLENGES



Cascading P2P Transactions

- Prosumer to Prosumer energy transactions -> **Cascading Transaction cost**
- CERC – ceiling trading margin



SoR CERC (Fixation of Trading Margin) Regulations, 2010

“16. Selling of power from the trader to trader is not prohibited under the Central Electricity Regulatory Commission (Procedure, Terms and Conditions for grant of trading licence and other related matters) Regulations, 2009. In such cases, there is a possibility of increase in price of the electricity traded in the market due to pan-caking. **In order to avoid cascading effect of the trading margin to be charged by various traders on the same transactions**, it has been decided that sum or the total of the trading margin charged by all the traders in the chain of transactions between generator and ultimate buyer i.e. to **say trading in case of multiple trader to trader transactions shall not exceed the ceiling trading margin specified under these regulations.**”

Pan-caking of P2P Transaction Charges Poses as a Risk

Will the Prosumer always find a buyer?

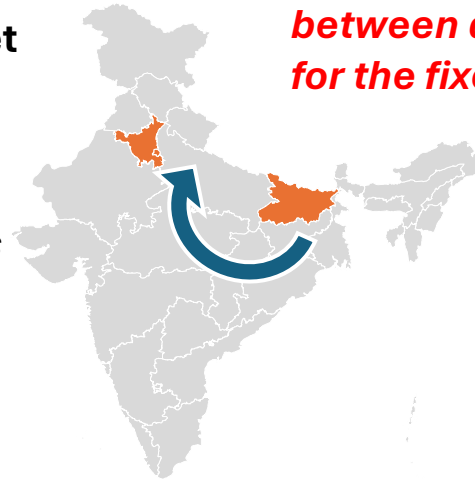
- If prosumer is not able to find a buyer – how will the excess energy be settled?
- Different States have different treatments
- For instance, in Bihar the Prosumer may not get anything

BERC (Rooftop Solar Grid Interactive Systems Based on Net & Gross Metering) Regulations, 2018

“8. Energy Accounting and Settlement

8.1 Net metering arrangement-

...Provided further that any excess injected electricity remains unadjusted at the end of the settlement period, **neither payment shall be made by the distribution licensee nor shall be carried forward to the next settlement period.**”



KERC (Implementation of Peer to Peer Solar Energy Transaction) Regulations 2024

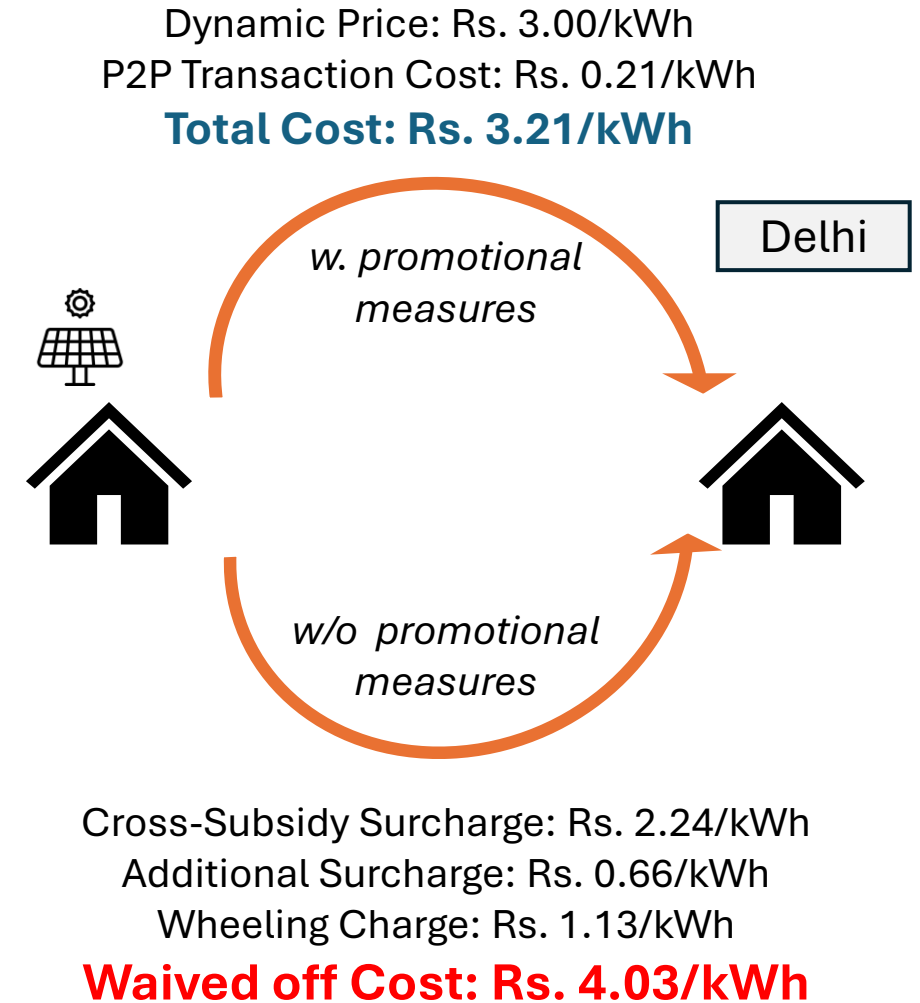
4. General Principles

(c) “Whenever there is no excess energy available in a P2P network, then the consumer’s energy needs will be met by distribution licensee. **Similarly, if no buyer is available for the P2P then it will be deemed to be sold back to distribution licensee as per Power Purchase Agreement executed between distribution licensee and the Consumer, for the fixed and guaranteed price.**”

Supply-Demand Mis-match May Lead to Prosumer Loss

Other Challenges

- **SUNSET CLAUSE**
 - Every promotional measure should eventually end
 - Market itself adopts
- **INTRA-DISTRIBUTION LOSS CALCULATION**
 - Different normative DL for DISCOMs
 - Which DL to be considered?
- **RISK OF BECOMING A COMMERCIAL ACTIVITY**
 - Instead of only surplus power, entire generation may be P2P
 - Original intent misconstrued
- **DISCOM SCHEDULING**



Way Forward

- **India Energy Stack – Rs. 51 Cr. fund** – pilots in Delhi, Gujarat, Andhra Pradesh, Uttar Pradesh & Maharashtra
- Incentives in the form of surcharge waivers should have a clear end date so as to provide proper **price signals**
- Can **Net-Metering & Open Access Regulations** be amended to include P2P?

Thank You!